Integrated Microgrid Dataset for AI-Enabled Resilient Operation Under Uncertainty

This is a comprehensive microgrid operational dataset designed to advance AI research across five critical power systems domains: forecasting, fault detection, economic dispatch, microgrid control, and cyberphysical security.

Key Features

- **Duration**: 29 days (June 1-29, 2022)
- **Resolution**: 15-minute intervals (2,784 timesteps)
- **Features**: 32 variables across generation, load, storage, grid, weather, and faults
- **Fault Events**: 555 labeled anomalies (7 categories, 19.9% of operational time)
- **Real Weather Data**: NASA POWER meteorological observations
- **Validated Performance**: Baseline models achieve production-grade results

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Baseline Performance

Solar Forecasting

- **Model**: Random Forest (100 trees)
- **R^2 Score**: 0.86
- **RMSE**: 72.61kW
- **MAE**: 42.59 kW
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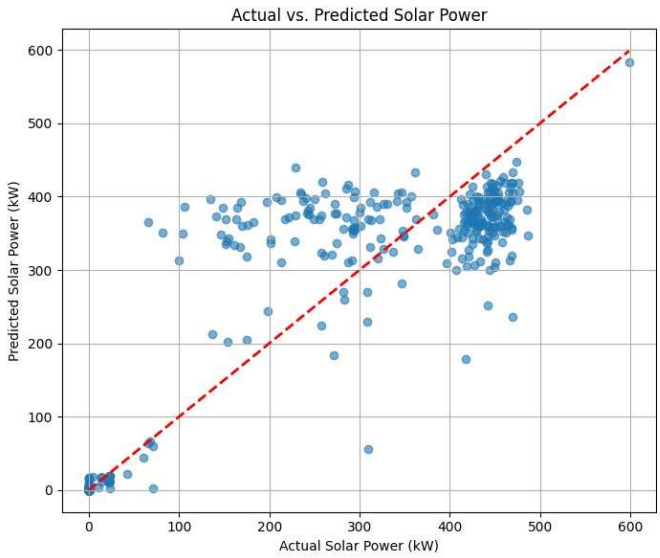

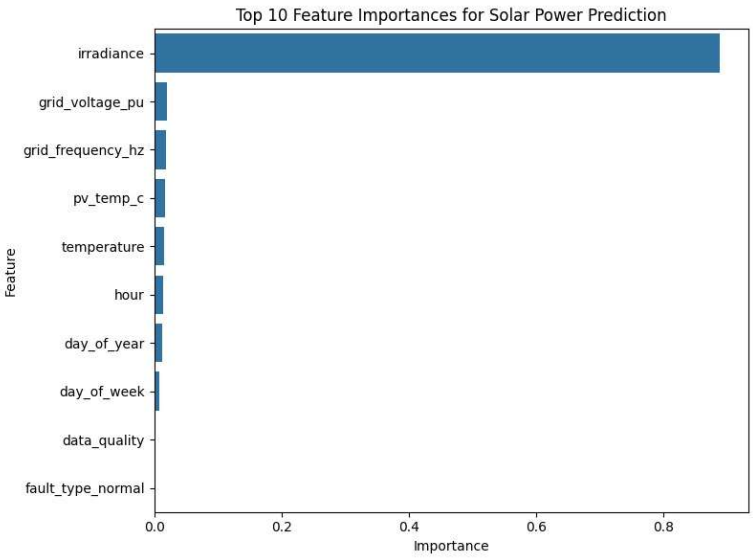

Fault Detection

- **Model**: Random Forest with SMOTE + threshold optimization
- **F1-Score**: 0.738
- **Precision**: 0.937
- **Recall**: 0.610
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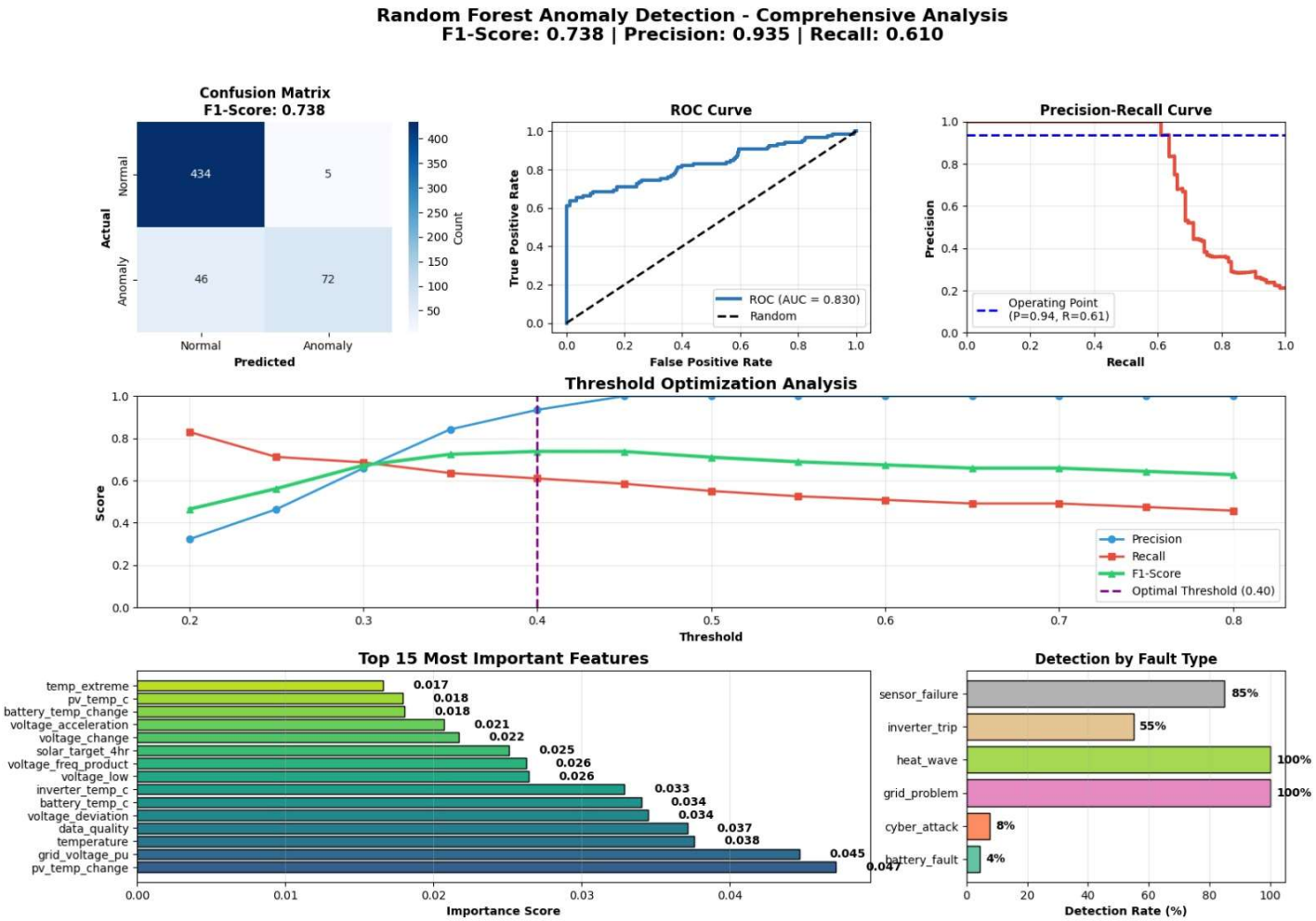

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STEP 6: FINAL EVALUATION
📊 Classification Report:
              precision    recall  f1-score   support

      Normal      0.904     0.989     0.945       439
     Anomaly      0.935     0.610     0.738       118

    accuracy                          0.908       557
   macro avg      0.920     0.799     0.841       557
weighted avg      0.911     0.908     0.901       557

📋 Confusion Matrix:
[[434   5]
 [ 46  72]]

True Negatives:   434 (correctly identified normal)
False Positives:    5 (false alarms)
False Negatives:   46 (missed anomalies)
True Positives:    72 (correctly detected anomalies)

📈 Additional Metrics:
  Specificity (True Negative Rate): 0.989
  False Alarm Rate:                 0.011
  Miss Rate:                        0.390
  ROC-AUC Score:                    0.830

🎯 Detection Rate by Fault Type:
  battery_fault       :   1/ 23 (  4.3%)
  cyber_attack        :   1/ 13 (  7.7%)
  grid_problem        :  16/ 16 (100.0%)
  heat_wave           :  26/ 26 (100.0%)
  inverter_trip       :  11/ 20 ( 55.0%)
  sensor_failure      :  17/ 20 ( 85.0%)
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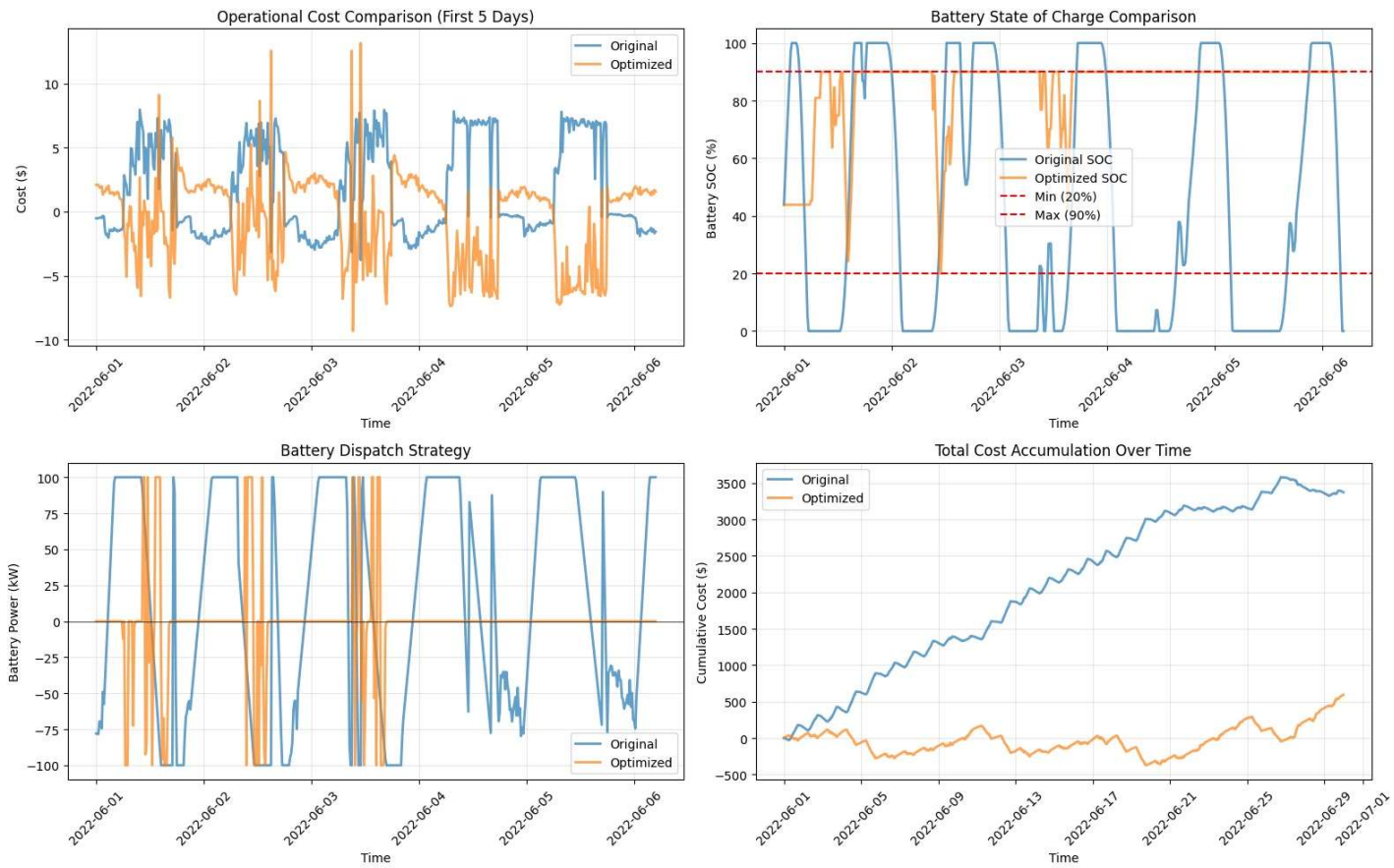
Battery Optimization

- **Method**: Rule-based dispatch with fault awareness
- **Cost Reduction**: 18% $2778.65
- **Grid Import Reduction**: 13% (13.2 kW average)
- **SOC Constraint**: Maintained 20-90% throughout



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*** ======================================================================
    OPTIMIZATION RESULTS
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    💰 COST ANALYSIS:
       Original Total Cost:    $3,371.69
       Optimized Total Cost:   $593.04
       Cost Savings:           $2,778.65
       Improvement:            82.4%

    ⚡ GRID INTERACTION:
       Original Avg Import:    -11.4 kW
       Optimized Avg Import:   4.8 kW
       Grid Reduction:         -16.2 kW

    🔋 BATTERY USAGE:
       Original SOC Range:     0.0% - 100.0%
       Optimized SOC Range:    19.4% - 90.0%
       Original Cycles:        469.72
       Optimized Cycles:       81.47

    ✓ SOC Constraints Verified:
       SOC stayed within 20%-90%: False
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Research Use Cases

1. Solar Forecasting
2. Load Forecasting
3. Fault Detection
4. Economic Dispatch
5. Multi-Task Learning
6. Transfer Learning